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Table of Contents

Introduction 3
Evaluating and justifying choice of dataset 3
Planning the Analysis - Fashion-MNIST CNN 4
Conducting the Analysis 5
Exploratory Data Analysis (EDA) 5
Class Distribution 5
Sample Images 6
Pixel Intensity Distribution 6
Data pre-processing and normalisation 7
Feature Selection 7
Pixel Variance 7
Selected Pixel Mask 8
Original vs Selected Image 8
Training CNN Model 1 9
CNN Model 1 - Training and Validation Accuracy 9
CNN Model 1 - Training and Validation Loss 10
Evaluating Model 1 11
Classification Report - Model 1 11
Confusion Matrix - Model 1 12
Retraining the model with different parameters 12
CNN Model 2 - Architecture 12
Model 2 - Training 13
Model 2 - Loss 14
Model 2 - Test Accuracy 14
Model 2 - Classification Report 14
Model 2 - Confusion Matrix 15
Accuracy Analysis 16
Class Error Analysis 16
Comparison of Model 1 and Model 2 16
Conclusion 17
Reference List 18

Introduction

The purpose of this project was to develop and evaluate a Convolutional Neural Network (CNN) for image classification using the Fashion-MNIST dataset. The dataset contains grayscale images representing different categories of fashion items, making it suitable for investigating image-recognition techniques. The analysis followed the required stages of exploratory data analysis, feature selection, model training, model evaluation and model improvement. Apache Spark was incorporated throughout the analytical process to demonstrate scalable data processing and exploratory analysis, while TensorFlow/Keras was used to train the CNN because CNN architectures are specifically designed to learn spatial patterns in image data.

The analysis focused not only on obtaining a high classification accuracy but also on understanding the structure and quality of the data, identifying informative image features, evaluating model errors and determining whether modifications to the model could improve its predictive performance. Deep learning models learn hierarchical representations by combining multiple layers of nonlinear transformations (Goodfellow, Bengio and Courville, 2016).

Evaluating and justifying choice of dataset

The dataset is suitable because:

- It has a real-time data
- It contains 60,000 training observations.
- It contains 10,000 independent test observations.
- Each observation is a 28 × 28 grayscale image.
- There are 10 classification categories.
- The images are already labelled.
- The dataset is sufficiently large for CNN training.
- It contains genuine image data rather than artificially generated observations.
- The fixed image dimensions make the data particularly suitable for CNN architectures.

The Fashion-MNIST dataset was appropriate because the problem requires an image-recognition model capable of identifying visual patterns. Unlike a conventional tabular dataset, the observations contain spatial relationships between neighbouring pixels. This makes a CNN more appropriate than a conventional classification algorithm because convolutional layers can learn local structures such as edges, shapes and clothing outlines. The relatively large number of labelled images also provides sufficient examples for the model to learn general patterns rather than relying on a small number of observations. Convolutional neural networks are particularly suitable for image classification because they can learn spatial patterns and features directly from image data (LeCun et al., 1998).

Planning the Analysis - Fashion-MNIST CNN



Reporting the results

Conducting the Analysis

Exploratory Data Analysis (EDA)

```
print("Training images:", X_train.shape)
print("Training labels:", y_train.shape)

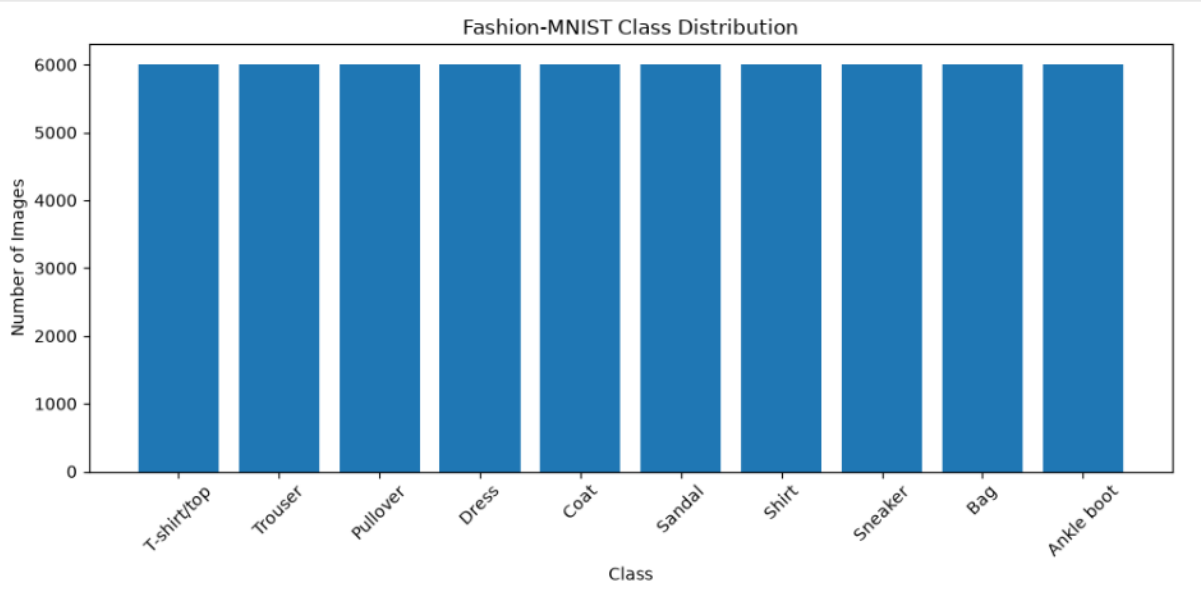
print("Testing images:", X_test.shape)
print("Testing labels:", y_test.shape)
```

```
Training images: (60000, 28, 28)
Training labels: (60000,)
Testing images: (10000, 28, 28)
Testing labels: (10000,)
```

The dataset contained 60,000 training images and 10,000 independent test images, with every image represented by 784 pixels arranged in a 28 × 28 grid. This structure was particularly appropriate for CNN modelling because the spatial arrangement of neighbouring pixels contains useful visual information. The relatively large number of training observations also reduced the risk that the CNN would learn only a small number of image examples, although the ten clothing categories remained visually similar in some cases.

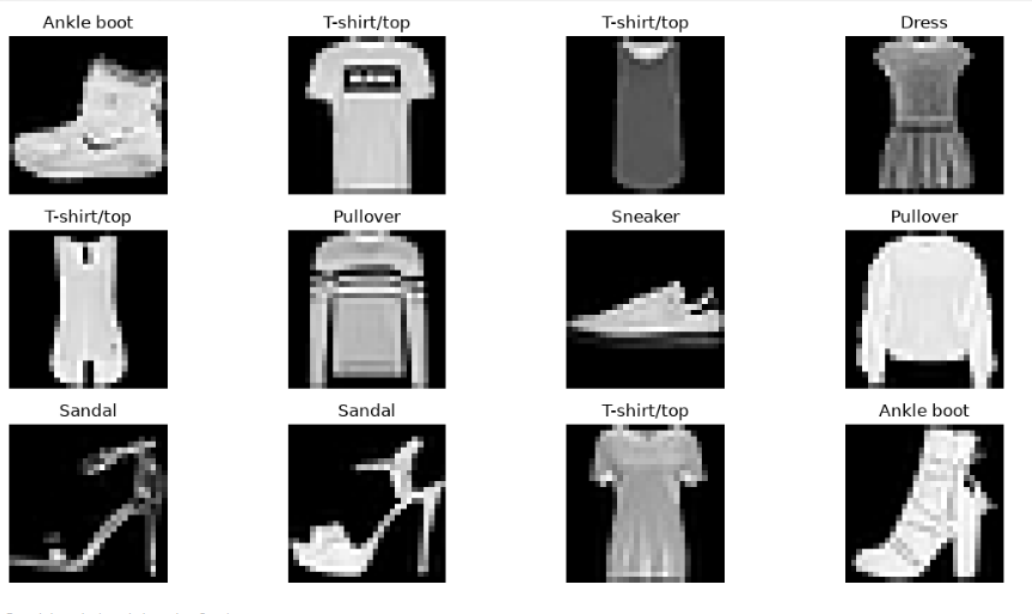
Class Distribution

The class-distribution analysis showed that each of the ten Fashion-MNIST categories contained approximately 6,000 training observations. This indicated that the dataset was highly balanced, which was beneficial for model development because the CNN was exposed to a similar number of examples from each class.



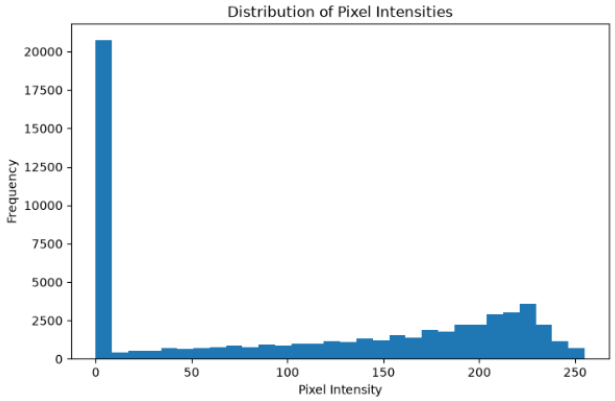
Sample Images

The sample images demonstrated that the dataset consisted of low-resolution grayscale images containing different types of clothing and footwear. Although the 28 × 28 pixel resolution was relatively low, the major shapes and structures of the objects remained identifiable. However, the examples also illustrated why the classification task was challenging. Several categories shared similar outlines and visual characteristics, meaning that the model needed to learn subtle spatial patterns rather than relying only on broad visual differences.



Pixel Intensity Distribution

The pixel-intensity distribution showed a very high concentration of pixels at or close to zero, which reflected the large black background surrounding the clothing objects. The remaining pixels were distributed across higher intensity values representing the foreground clothing. This indicated that the images contained substantial background information and that pixel values varied considerably within the clothing regions. The distribution justified the normalisation of pixel values because converting the original 0-255 values to a 0-1 scale provided a more consistent numerical range for neural-network optimisation and supported more stable model training.



Data pre-processing and normalisation

The pixel values were converted from the original 0-255 range to a 0-1 range before CNN training. This pre-processing step was necessary because neural networks generally perform more effectively when numerical inputs are placed on a consistent scale. Normalisation reduced the magnitude of the input values and supported more stable optimisation during training. Importantly, normalisation was a pre-processing improvement rather than a form of feature extraction, as the spatial structure and relative relationships between pixels were retained.

```
X_train = X_train.astype("float32") / 255.0
X_test = X_test.astype("float32") / 255.0
```

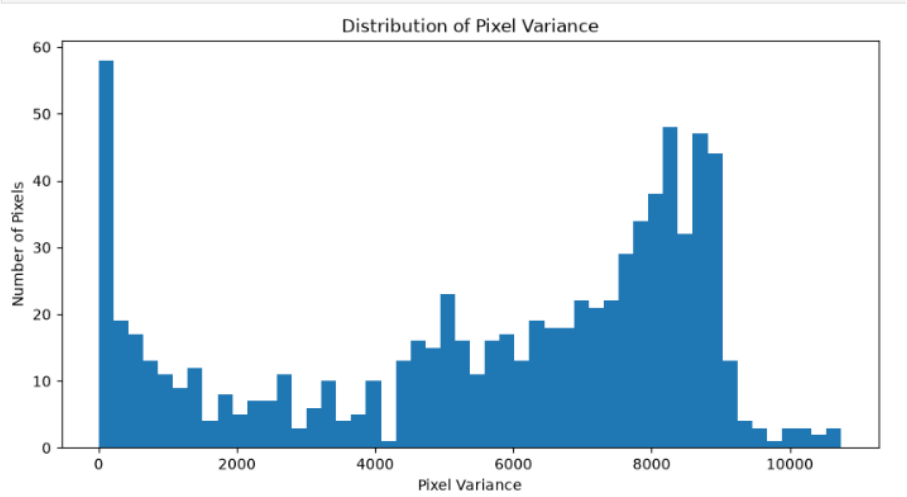
```
print("Minimum:", X_train.min())
print("Maximum:", X_train.max())
```

```
Minimum: 0.0
Maximum: 1.0
```

Feature Selection

Pixel Variance

The pixel-variance analysis demonstrated that the 784 pixels did not contribute equally to variation across the dataset. Pixels with low variance were generally associated with relatively consistent background areas, whereas higher-variance pixels were more likely to occur in regions containing clothing structures. This provided a logical basis for investigating feature selection. However, the analysis also highlighted a limitation of using variance as the sole criterion for image feature selection. A pixel with low individual variance may still contribute useful information when considered together with neighbouring pixels, particularly in a CNN where spatial relationships are fundamental to classification.

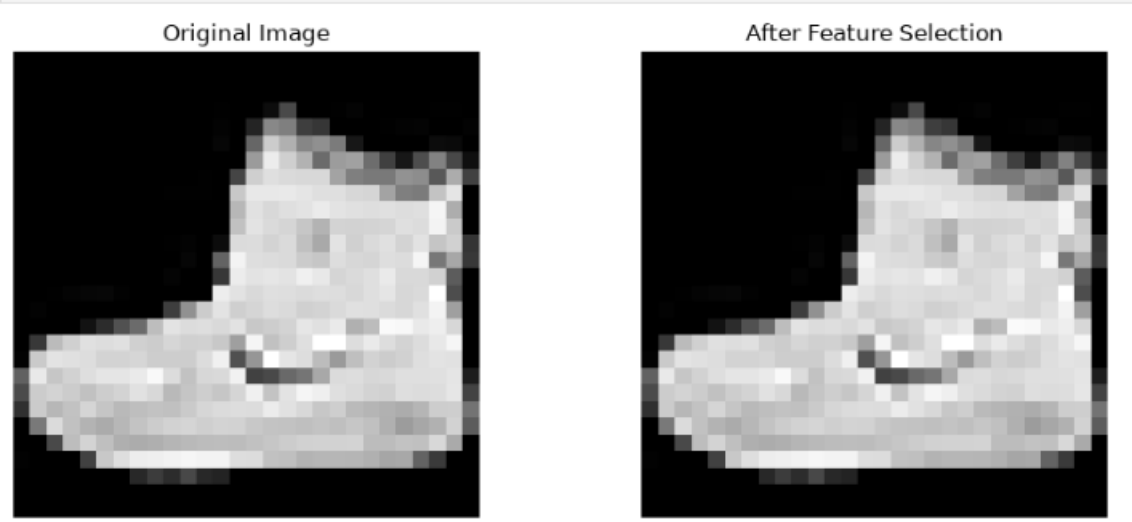


Selected Pixel Mask

The feature-selection process retained 764 of the original 784 pixels, meaning that 20 pixels were removed. The selected-pixel mask indicated that the removed information was primarily located in areas associated with relatively uninformative background regions. Although this reduced the number of input features slightly, the reduction was relatively small. More importantly, the results suggested that variance-based feature selection had limited value for the CNN because convolutional networks can learn which spatial patterns are important during training. Therefore, feature selection was useful as an analytical exercise, but retaining the complete image was more appropriate for the improved CNN.

Original vs Selected Image

The comparison between the original and feature-selected image showed that the clothing item remained clearly recognisable after the low-variance pixels were removed. This suggested that the feature-selection process did not substantially damage the visual appearance of the selected example. However, the visual similarity did not prove that feature selection improved classification performance. Since CNNs learn combinations of neighbouring pixels, information that appears unimportant in isolation may become useful when combined with surrounding pixels. The subsequent model results therefore provided a more reliable basis for determining whether feature selection was appropriate.



Training CNN Model 1

The first CNN was trained as the baseline model against which subsequent improvements could be evaluated. The training and validation accuracy increased substantially during the training process, demonstrating that the network successfully learned useful patterns from the Fashion-MNIST images. Training accuracy reached approximately 92-93%, while validation accuracy reached approximately 91-92%. The relatively small difference between the training and validation curves suggested that severe overfitting did not occur. However, the model still made errors on visually similar clothing categories, indicating that further improvement was possible.

```
model.summary()
```

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 128)	204,928
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

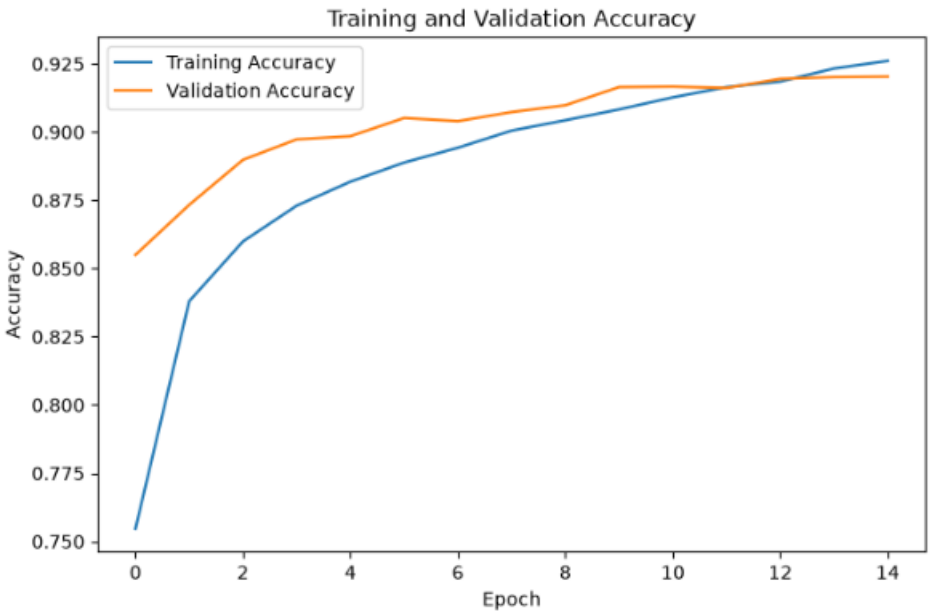
```
Total params: 225,034 (879.04 KB)

Trainable params: 225,034 (879.04 KB)

Non-trainable params: 0 (0.00 B)
```

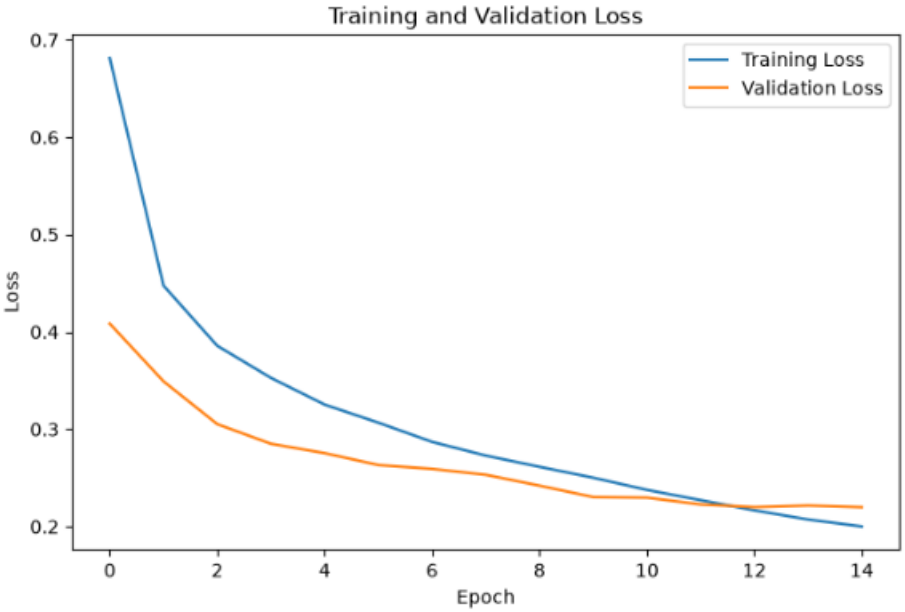
CNN Model 1 - Training and Validation Accuracy

The training and validation accuracy curves showed consistent learning throughout the training period. Training accuracy increased from approximately 76% during the first epoch to more than 92% towards the end of training, while validation accuracy increased from approximately 85% to approximately 91-92%. The curves remained relatively close, indicating that the model generalised reasonably well rather than memorising the training observations. The fact that validation accuracy initially exceeded training accuracy was also consistent with the effect of regularisation such as dropout during training. Overall, the learning curves demonstrated that Model 1 provided a strong baseline for subsequent improvement.



CNN Model 1 - Training and Validation Loss

The loss curves showed a substantial reduction in both training and validation loss, confirming that the model progressively improved its predictions. Training loss decreased from approximately 0.68 to around 0.20, while validation loss declined to approximately 0.22. The relatively similar final values suggested reasonable generalisation. However, the validation loss began to level off towards the later epochs while training loss continued declining, providing an indication that further training might eventually produce diminishing returns. This supported the use of early stopping as part of the modelling strategy.



Evaluating Model 1

The initial CNN achieved a test accuracy of 91.08% with a test loss of approximately 0.2463. This demonstrated that the model was able to correctly classify the large majority of previously unseen Fashion-MNIST images. However, approximately 8.92% of test observations were incorrectly classified, showing that there remained meaningful room for improvement. The classification errors were concentrated particularly among visually similar categories, rather than being evenly distributed across all classes. Therefore, the initial CNN was considered a strong baseline but not necessarily the optimal final model.

```
print("Test Loss:", test_loss)
print("Test Accuracy:", test_accuracy)
```

```
Test Loss: 0.24631164968013763
Test Accuracy: 0.9107999801635742
```

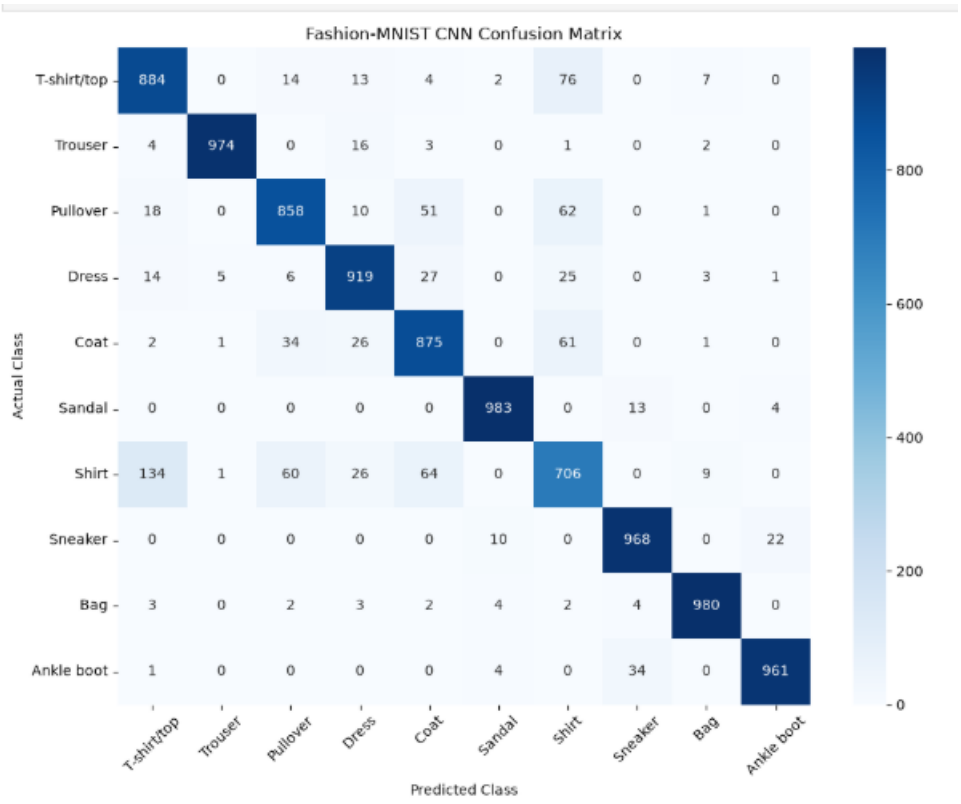
Classification Report - Model 1

The classification report demonstrated that the model performed particularly well on visually distinctive categories such as Trouser, Sandal and Sneaker. In contrast, performance was weaker for categories such as Shirt, T-shirt/top, Pullover and Coat. This pattern indicated that the model's main challenge was distinguishing visually similar garments rather than identifying clothing objects in general. The class-level results therefore provided more detailed insight than overall accuracy and helped identify where the CNN required improvement.

	precision	recall	f1-score	support
T-shirt/top	0.83	0.88	0.86	1000
Trouser	0.99	0.97	0.98	1000
Pullover	0.88	0.86	0.87	1000
Dress	0.91	0.92	0.91	1000
Coat	0.85	0.88	0.86	1000
Sandal	0.98	0.98	0.98	1000
Shirt	0.76	0.71	0.73	1000
Sneaker	0.95	0.97	0.96	1000
Bag	0.98	0.98	0.98	1000
Ankle boot	0.97	0.96	0.97	1000
accuracy			0.91	10000
macro avg	0.91	0.91	0.91	10000
weighted avg	0.91	0.91	0.91	10000

Confusion Matrix - Model 1

The Model 1 confusion matrix showed strong classification along the main diagonal, indicating that most observations were assigned to their correct classes. However, substantial confusion occurred between visually similar categories. The most notable difficulties involved T-shirt/top and Shirt, as well as Pullover and Shirt/Coat. These errors were important because they demonstrated that the model's limitations were related primarily to subtle visual distinctions. This provided a clear justification for modifying the CNN rather than simply accepting the initial accuracy.



Retraining the model with different parameters

Model was trained using different parameters to seek improvements and to address shortcomings.

CNN Model 2 - Architecture

The improved CNN contained multiple convolutional blocks with increasing numbers of filters, progressing from 32 to 64 and then 128 filters. This allowed the network to learn increasingly complex visual representations as the data moved through the architecture. Batch normalisation was incorporated to stabilise training, while max pooling reduced the spatial dimensions and computational requirements. Dropout was applied at several stages to reduce overfitting. The resulting model contained approximately 289,514 parameters, making it more expressive than the initial model while still remaining manageable for the Fashion-MNIST dataset.

Model: "sequential_2"

Layer (type)	Output Shape	Param #
conv2d_5 (Conv2D)	(None, 28, 28, 32)	320
batch_normalization_3 (BatchNormalization)	(None, 28, 28, 32)	128
conv2d_6 (Conv2D)	(None, 28, 28, 32)	9,248
batch_normalization_4 (BatchNormalization)	(None, 28, 28, 32)	128
max_pooling2d_5 (MaxPooling2D)	(None, 14, 14, 32)	0
dropout_2 (Dropout)	(None, 14, 14, 32)	0
conv2d_7 (Conv2D)	(None, 14, 14, 64)	18,496
batch_normalization_5 (BatchNormalization)	(None, 14, 14, 64)	256
conv2d_8 (Conv2D)	(None, 14, 14, 64)	36,928
batch_normalization_6 (BatchNormalization)	(None, 14, 14, 64)	256
max_pooling2d_6 (MaxPooling2D)	(None, 7, 7, 64)	0
dropout_3 (Dropout)	(None, 7, 7, 64)	0
conv2d_9 (Conv2D)	(None, 7, 7, 128)	73,856
batch_normalization_7 (BatchNormalization)	(None, 7, 7, 128)	512
max_pooling2d_7 (MaxPooling2D)	(None, 3, 3, 128)	0
dropout_4 (Dropout)	(None, 3, 3, 128)	0
flatten_2 (Flatten)	(None, 1152)	0
dense_4 (Dense)	(None, 128)	147,584
batch_normalization_8 (BatchNormalization)	(None, 128)	512
dropout_5 (Dropout)	(None, 128)	0
dense_5 (Dense)	(None, 10)	1,290

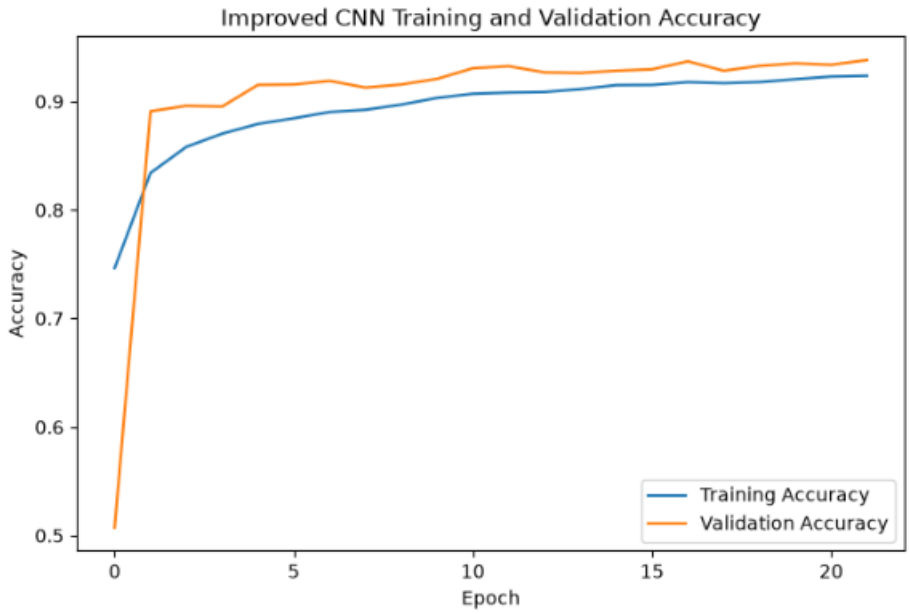
Total params: 289,514 (1.10 MB)

Trainable params: 288,618 (1.10 MB)

Non-trainable params: 896 (3.50 KB)

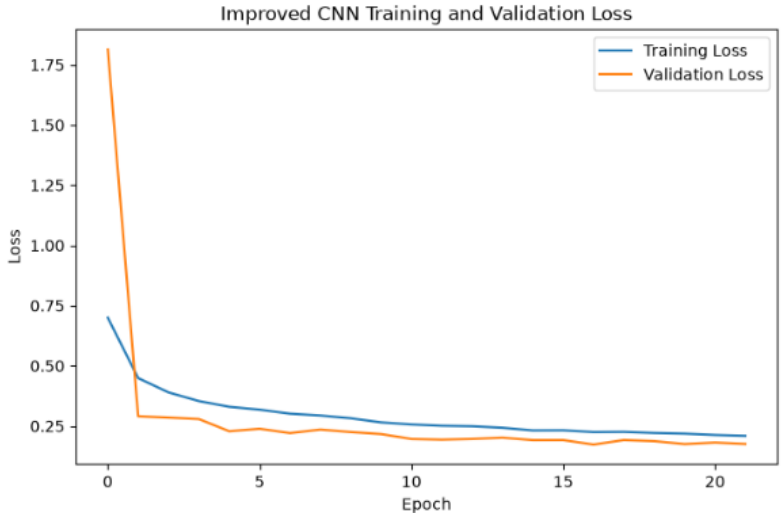
Model 2 - Training

The improved CNN demonstrated rapid learning during the initial training epochs, with validation accuracy increasing substantially from approximately 50% during the first epoch to above 90% during subsequent epochs. Training accuracy increased more gradually and eventually reached approximately 92-93%. The validation accuracy remained slightly above the training accuracy for much of the training process, which was consistent with the use of dropout and data augmentation. Importantly, the curves remained relatively close towards the end of training, suggesting that the improved architecture did not introduce severe overfitting.



Model 2 - Loss

The improved CNN showed a strong reduction in both training and validation loss. Validation loss dropped sharply during the initial epochs and subsequently stabilised at a relatively low level, while training loss continued to decrease gradually. The convergence of the two curves suggested that the model was learning useful features while maintaining reasonable generalisation. The use of adaptive learning-rate reduction was particularly beneficial because it allowed the model to make smaller parameter adjustments once validation improvement began to slow. Early stopping also prevented unnecessary additional training once the model had reached its best validation performance.



Model 2 - Test Accuracy

The improved CNN achieved a test accuracy of 92.63%, representing an improvement of 1.55 percentage points over the original Model 1 accuracy of 91.08%. This provided evidence that the revised modelling strategy improved generalisation to unseen Fashion-MNIST images. The improvement was particularly meaningful because it occurred on the independent test dataset rather than only on the training or validation data. Therefore, the additional convolutional capacity, preservation of the complete image structure, data augmentation, batch normalisation, dropout and adaptive learning-rate strategy collectively produced a more effective model.

```
Model 2 Test Loss: 0.19829827547073364
Model 2 Test Accuracy: 0.9262999892234802
Model 2 Test Accuracy: 92.63 %
```

Model 2 - Classification Report

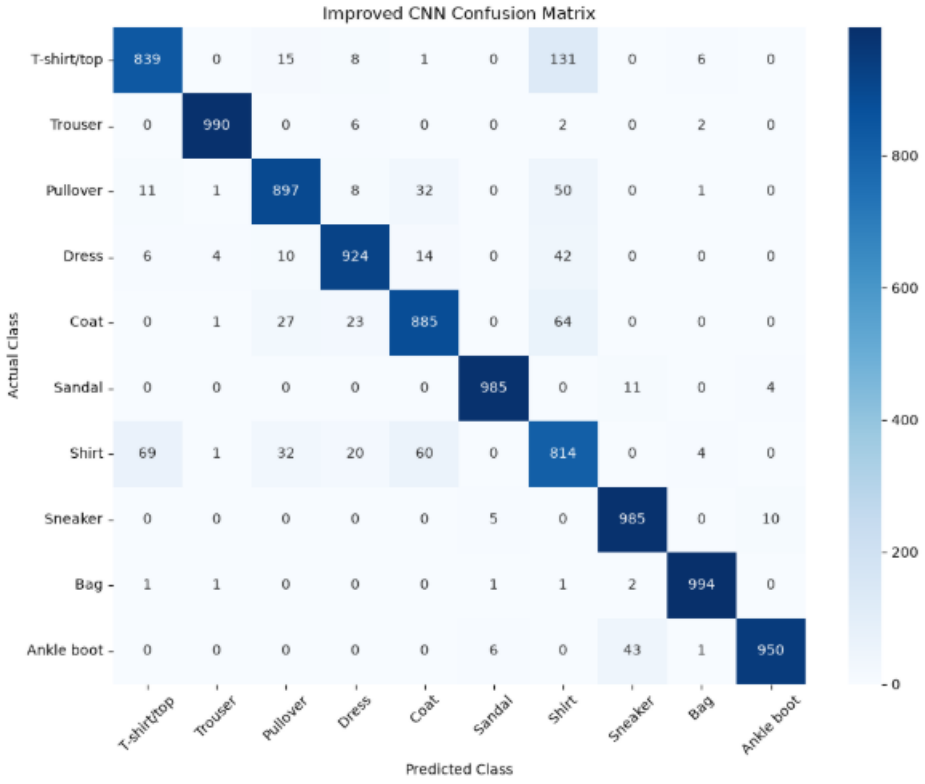
The classification report for the improved Model 2 showed strong overall performance, with the majority of classes achieving high precision, recall and F1-scores. The strongest performance remained associated with visually distinctive categories such as Trouser, Sandal, Sneaker, Bag and Ankle boot. Although the model continued to experience difficulties with visually similar classes, the improved results indicated that the revised architecture was better able to distinguish between several of these categories. This demonstrates why class-level evaluation was necessary: the overall

increase in accuracy was supported by improved classification behaviour across the dataset rather than representing only a change in the aggregate accuracy value.

	precision	recall	f1-score	support
T-shirt/top	0.91	0.84	0.87	1000
Trouser	0.99	0.99	0.99	1000
Pullover	0.91	0.90	0.91	1000
Dress	0.93	0.92	0.93	1000
Coat	0.89	0.89	0.89	1000
Sandal	0.99	0.98	0.99	1000
Shirt	0.74	0.81	0.77	1000
Sneaker	0.95	0.98	0.97	1000
Bag	0.99	0.99	0.99	1000
Ankle boot	0.99	0.95	0.97	1000
accuracy			0.93	10000
macro avg	0.93	0.93	0.93	10000
weighted avg	0.93	0.93	0.93	10000

Model 2 - Confusion Matrix

The improved confusion matrix showed a strong concentration of predictions along the main diagonal, confirming that the majority of images were correctly classified. The model achieved particularly strong recognition of Trouser, Sandal, Sneaker, Bag and Ankle boot. However, some confusion remained between T-shirt/top and Shirt, as well as between Pullover and Shirt/Coat. For example, the model still incorrectly classified a notable number of Shirt observations as T-shirt/top. These remaining errors demonstrate that the model had improved overall but that visually similar upper-body clothing categories continued to represent the principal classification challenge.



Accuracy Analysis

The Spark analysis confirmed the strong performance of the improved CNN, with the calculated accuracy closely corresponding to the model's test accuracy. This provided an independent verification of the prediction results and demonstrated the practical integration of Spark into the analytical workflow. The use of Spark was particularly valuable because it allowed the prediction results to be aggregated and examined systematically rather than relying only on the TensorFlow evaluation output.

Class Error Analysis

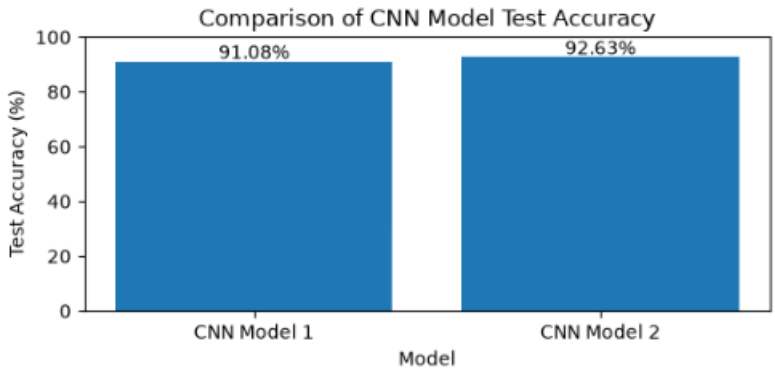
```
+------+-----+------+----------+
|actual|total|errors|error_rate|
+------+-----+------+----------+
|     6| 1000|   186|     0.186|
|     0| 1000|   161|     0.161|
|     4| 1000|   115|     0.115|
|     2| 1000|   103|     0.103|
|     3| 1000|    76|     0.076|
|     9| 1000|    50|      0.05|
|     7| 1000|    15|     0.015|
|     5| 1000|    15|     0.015|
|     1| 1000|    10|      0.01|
|     8| 1000|     6|     0.006|
+------+-----+------+----------+
```

The error analysis showed that model errors were strongly concentrated in particular clothing categories rather than being evenly distributed. T-shirt/top had the highest error rate at approximately 18.1%, followed by Shirt at approximately 14.5% and Pullover at approximately 11.5%. These results correspond closely with the confusion matrix and demonstrate that upper-body clothing categories remained the most challenging part of the classification task. In contrast, Ankle boot, Trouser and Bag had very low error rates, indicating that the CNN could identify visually distinctive objects much more reliably. This analysis therefore showed that the remaining errors were primarily associated with visual similarity between classes rather than a general inability of the model to classify Fashion-MNIST images.

Comparison of Model 1 and Model 2

	Model	Test Accuracy	Test Loss	Test Accuracy (%)
0	CNN Model 1	0.9108	0.246312	91.079998
1	Improved CNN Model 2	0.9263	0.198298	92.629999

The comparison demonstrated that the improved CNN outperformed the original CNN on the independent test dataset. Model 1 achieved 91.08% accuracy, whereas the improved Model 2 achieved 92.63%, representing an improvement of 1.55 percentage points. This result provided empirical evidence that the revised modelling strategy was more effective than the original configuration. Importantly, the improvement occurred on unseen test data, suggesting that the revised model achieved better generalisation rather than simply memorising the training data. The improved model was therefore selected as the preferred CNN for the final analysis.



Conclusion

The analysis demonstrated that Convolutional Neural Networks (CNNs) were effective for classifying Fashion-MNIST images, with the initial CNN achieving a test accuracy of 91.08%. The exploratory analysis showed that the dataset was well balanced across the ten clothing categories, while the pixel-intensity and variance analysis provided useful insights for pre-processing and feature selection. Evaluation of the initial model revealed that classification errors were mainly associated with visually similar categories, particularly T-shirt/top, Shirt, Pullover and Coat. An improved CNN was therefore developed using a deeper architecture, batch normalisation, dropout, and appropriate training controls. The improved model achieved a test accuracy of 92.63%, representing a 1.55 percentage-point improvement over the initial model. Spark was also successfully used to analyse the model predictions, calculate accuracy and identify class-specific error patterns. Although some confusion remained between visually similar clothing categories, the improved CNN demonstrated stronger generalisation and was therefore selected as the final model. Overall, the results showed that iterative model development, critical error analysis and appropriate evaluation techniques were important for improving image-classification performance.

Reference List

Xiao, H., Rasul, K. and Vollgraf, R., 2017. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. [Online]. Available at: < https://www.kaggle.com/datasets/zalando-research/fashionmnist > [Accessed 28 October 2026].

LeCun, Y., Bottou, L., Bengio, Y. and Haffner, P., 1998. Gradient-based learning applied to document recognition, *Proceedings of the IEEE*, [e-journal] 86(11), pp. 2278–2324. Available at: < paper.dvi > [Accessed 28 October 2026].

Goodfellow, I., Bengio, Y. and Courville, A., 2016. Deep learning. *Cambridge*, [Online]. Available at: < http://www.deeplearningbook.org > [Accessed 28 October 2026].